MultiQT notes

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1 Studentized score

Writing the score as a signed residual gives the usual additive MultiQT correction. Studentizing that residual lead to multiplicatively scaled offsets in a MulitQT update or offsets in a scaled coordinate system. With changing scale, the accumulated correction as well as the new offset are scaled. Thus, a positive offset that was built up in a underprediction/ upswing phase can easily be removed after the peak (so the idea). With quantile specific scales the ordering constraint lead to the feasible set being a diagonally transformed shifted isotonic cone. This is still an isotnic cone and a closed convex polyhedron with a unique Euclidean projection (weighted isotonic regression). The gradient steps now optimize weighted quantile loss. The coverage criterion and calibration target are unchanged (strictly increasing invertible transformation).

For the standard MulitQT, [**?**], et $\mathcal{A} = \{\alpha_1 < \cdots < \alpha_m\} \subset (0,1)$ and let $b_t = (b_t^\alpha)_{\alpha\in\mathcal{A}}$ be the base quantile forecasts available before observing y_t. Define

$$\mathcal{K} = \{q \in \mathbb{R}^m : q^{\alpha_1} \le \cdots \le q^{\alpha_m}\}, \qquad \Pi_C(x) = \arg\min_{z\in C} \|z - x\|^2,$$

where C is a nonempty closed convex set. Assume $b_t \in \mathcal{K}$. The original MultiQT procedure is

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t^{\text{std}} = b_t + \theta_t, \tag{1}$$

$$\text{cov}_t^{\text{std},\alpha} = \mathbf{1}\{y_t \le q_t^{\text{std},\alpha}\}, \qquad \tilde{\theta}_{t+1}^\alpha = \tilde{\theta}_t^\alpha - \eta(\text{cov}_t^{\text{std},\alpha} - \alpha). \tag{2}$$

Here $\tilde{\theta}_t$ is the hidden offset and θ_t is the played (projected) offset. Both have outcome units.

We can rewrite this update procedure in the score-threshold update by [**?**] with the resiudal as score:

A score function $S_t(y)$ maps a candidate outcome to a real number. Let $r_t = S_t(y_t)$ be the observed score and c_t its issued threshold. The corresponding prediction set is

$$\mathcal{P}_t = \{y : S_t(y) \le c_t\}, \qquad \text{err}_t = \mathbf{1}\{r_t > c_t\}. \tag{3}$$

. Then update according to

$$\begin{aligned} c_{t+1} &= c_t + \eta(\text{err}_t - \delta) \\ &= c_t - \eta(\mathbf{1}\{r_t \le c_t\} - \alpha). \end{aligned} \tag{4}$$

The second line follows from $\text{err}_t = 1 - \mathbf{1}\{r_t \le c_t\}$ and setting $\delta = 1 - \alpha$. The threshold c_t is measured in score units and determines a prediction set through score inversion. For the signed affine scores below, that set's endpoint is the outcome quantile forecast.

With quantile loss and holding the observed score fixed:

$$\rho_\alpha(z, r) = \alpha(r - z)_+ + (1 - \alpha)(z - r)_+. \tag{5}$$

$$\partial_z \rho_\alpha(z, r) = \begin{cases} \{-\alpha\}, & z < r, \\ [-\alpha, 1-\alpha], & z = r, \\ \{1-\alpha\}, & z > r. \end{cases} \tag{6}$$

Selecting $1-\alpha$ at equality gives

$$g_\alpha(z; r) = \mathbf{1}\{r \le z\} - \alpha \ \in \ \partial_z \rho_\alpha(z, r). \tag{7}$$

Thus $c_{t+1} = c_t - \eta g_\alpha(c_t; r_t)$ is (4). The score's distance from the threshold affects the loss; the side of the threshold determines the update direction and size.

For each level define the score and its observed value by

$$S_t^{\mathrm{std},\alpha}(y) = y - b_t^\alpha, \qquad r_t^{\mathrm{std},\alpha} = y_t - b_t^\alpha. \tag{8}$$

Using θ_t^α as the score threshold gives

$$\{y : S_t^{\mathrm{std},\alpha}(y) \le \theta_t^\alpha\} = (-\infty, b_t^\alpha + \theta_t^\alpha] = (-\infty, q_t^{\mathrm{std},\alpha}]. \tag{9}$$

Consequently,

$$\mathrm{cov}_t^{\mathrm{std},\alpha} = \mathbf{1}\{y_t \le b_t^\alpha + \theta_t^\alpha\} = \mathbf{1}\{r_t^{\mathrm{std},\alpha} \le \theta_t^\alpha\}. \tag{10}$$

and substitution into (2) gives

$$\boxed{\tilde{\theta}_{t+1}^\alpha = \tilde{\theta}_t^\alpha - \eta(\mathbf{1}\{r_t^{\mathrm{std},\alpha} \le \theta_t^\alpha\} - \alpha).} \tag{11}$$

The original MultiQT loss has the exact residual-coordinate representation

$$\begin{aligned} \ell_t^{\mathrm{std}}(\theta) &= \sum_{\alpha \in \mathcal{A}} \rho_\alpha(b_t^\alpha + \theta^\alpha, y_t) \\ &= \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\theta^\alpha, y_t - b_t^\alpha) = \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\theta^\alpha, r_t^{\mathrm{std},\alpha}). \end{aligned} \tag{12}$$

The threshold c_t of the scalar score tracker corresponds to the offset θ_t^α when the score is $y - b_t^\alpha$. For a single level the constraint is all of $\mathbb{R}$, so the hidden and played states coincide and (11) is exactly scalar quantile tracking. For multiple levels, MultiQT uses a joint projection: evaluate g_α at the played θ_t^α, but apply the increment to the hidden $\tilde{\theta}_t^\alpha$.

Now replace the residual as score by the studentized residual:

Choose finite scales $\sigma_t^\alpha > 0$ from information available at issuance. They are fixed when the threshold is optimized. Define

$$S_t^{\mathrm{stu},\alpha}(y) = \frac{y - b_t^\alpha}{\sigma_t^\alpha}, \qquad r_t^{\mathrm{stu},\alpha} = \frac{y_t - b_t^\alpha}{\sigma_t^\alpha}. \tag{13}$$

Use $\tilde{u}_t$ and u_t for hidden and played thresholds in studentized units. Inverting the score gives

$$\begin{aligned} S_t^{\mathrm{stu},\alpha}(y) \le u_t^\alpha \quad &\Longleftrightarrow \quad y \le b_t^\alpha + \sigma_t^\alpha u_t^\alpha, \\ q_t^{\mathrm{stu},\alpha} = b_t^\alpha + \sigma_t^\alpha u_t^\alpha, \qquad & \mathrm{cov}_t^{\mathrm{stu},\alpha} = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\}. \end{aligned} \tag{14}$$

Standard MultiQT tracks the outcome-unit residual threshold θ_t^α and issues $b_t^\alpha + \theta_t^\alpha$. Studentization tracks u_t^α and issues $b_t^\alpha + \sigma_t^\alpha u_t^\alpha$. The factor σ_t^α enters through the inverse score map.

Apply the same quantile-loss construction to the studentized scores:

$$\begin{aligned} \ell_t^{\mathrm{stu}}(u) &= \sum_{\alpha \in \mathcal{A}} \rho_\alpha(u^\alpha, r_t^{\mathrm{stu},\alpha}) \\ &= \sum_{\alpha \in \mathcal{A}} \frac{1}{\sigma_t^\alpha} \rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t). \end{aligned} \tag{15}$$

Using (7), its selected subgradient is

$$g_t^{\mathrm{stu},\alpha}(u_t) = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\} - \alpha = \mathrm{cov}_t^{\mathrm{stu},\alpha} - \alpha. \tag{16}$$

The chain rule gives the same result. Away from ties, for one level,

$$\begin{aligned}\frac{\partial}{\partial u^\alpha}\left[\frac{\rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t)}{\sigma_t^\alpha}\right] &= \frac{1}{\sigma_t^\alpha}(\mathbf{1}\{y_t \le b_t^\alpha + \sigma_t^\alpha u^\alpha\} - \alpha)\sigma_t^\alpha \\ &= \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u^\alpha\} - \alpha. \end{aligned} \tag{17}$$

At a tie use the selected subgradient in (7). The inverse scale in the loss cancels the scale in the inverse score map.

Retaining MultiQT's lazy update therefore gives

$$\boxed{\tilde{u}_{t+1}^\alpha = \tilde{u}_t^\alpha - \eta g_t^{\mathrm{stu},\alpha}(u_t) = \tilde{u}_t^\alpha - \eta\left(\mathbf{1}\left\{\frac{y_t - b_t^\alpha}{\sigma_t^\alpha} \le u_t^\alpha\right\} - \alpha\right).} \tag{18}$$

Initialize $\tilde{u}_1 = 0$. The score is compared with the *played* threshold; the increment is applied to the *hidden* threshold. The next section specifies the projection that produces u_t.

	Standard MultiQT	Studentized extension
Observed score	$r_t^{\mathrm{std},\alpha} = y_t - b_t^\alpha$	$r_t^{\mathrm{stu},\alpha} = (y_t - b_t^\alpha)/\sigma_t^\alpha$
Gradient	$\mathbf{1}\{r_t^{\mathrm{std},\alpha} \le \theta_t^\alpha\} - \alpha$	$\mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\} - \alpha$
Update origin	$\tilde{\theta}_t^\alpha$	$\tilde{u}_t^\alpha$
Issued forecast	$b_t^\alpha + \theta_t^\alpha$	$b_t^\alpha + \sigma_t^\alpha u_t^\alpha$

The gradient has the same form because both losses apply quantile loss to a score threshold. The observed scores, state units, and issued forecasts change. An unnormalized outcome-unit loss would instead have gradient $\sigma_t^\alpha(\mathrm{cov}_t^{\mathrm{stu},\alpha} - \alpha)$ with respect to u^α, and would define a different update.

Now derive the monotone constraints or feasible space for projection:

The constraint always concerns the outcome forecasts: $q_t^{\alpha_1} \le \cdots \le q_t^{\alpha_m}$. For the ordinary residual score, this requires

$$b_t + \theta \in \mathcal{K} \iff \theta \in C_t^{\mathrm{std}} := \mathcal{K} - b_t, \tag{19}$$

$$\theta^{\alpha_i} - \theta^{\alpha_{i+1}} \le b_t^{\alpha_{i+1}} - b_t^{\alpha_i}, \qquad i = 1, \ldots, m-1. \tag{20}$$

For studentization, set

$$A_t = \mathrm{diag}(\sigma_t^{\alpha_1}, \ldots, \sigma_t^{\alpha_m}), \qquad q_t^{\mathrm{stu}} = b_t + A_t u_t.$$

Then

$$\boxed{C_t^{\mathrm{stu}} = \{u \in \mathbb{R}^m : b_t + A_t u \in \mathcal{K}\} = A_t^{-1}(\mathcal{K} - b_t).} \tag{21}$$

This is a closed convex set given by linear inequalities. It contains zero when $b_t \in \mathcal{K}$. In general $A_t^{-1}\mathcal{K} \ne \mathcal{K}$, so it cannot be simplified to $\mathcal{K} - A_t^{-1}b_t$ for unequal scales. Compared to original MultiQT with residual constraint $\theta + b_t \in \mathcal{K}$, we now have $A_t u + b_t \in \mathcal{K}$. But Euclidean projection is feasible.

$$u_t = \Pi_{C_t^{\mathrm{stu}}}(\tilde{u}_t), \qquad q_t^{\mathrm{stu}} = b_t + A_t u_t. \tag{22}$$

Substituting $u = A_t^{-1}(q - b_t)$ into the projection objective yields

$$\begin{aligned} q_t^{\text{stu}} &= \arg\min_{q \in \mathcal{K}} \|A_t^{-1}(q - b_t) - \tilde{u}_t\|^2 \\ &= \arg\min_{q \in \mathcal{K}} \sum_{\alpha \in \mathcal{A}} \frac{(q^\alpha - b_t^\alpha - \sigma_t^\alpha \tilde{u}_t^\alpha)^2}{(\sigma_t^\alpha)^2}. \end{aligned} \tag{23}$$

Define $\Pi_{\mathcal{K}}^W(v) = \arg\min_{q \in \mathcal{K}} (q - v)^\top W (q - v)$ for a positive-definite matrix W. The exact identity is

$$\boxed{b_t + A_t \Pi_{A_t^{-1}(\mathcal{K} - b_t)}(\tilde{u}_t) = \Pi_{\mathcal{K}}^{A_t^{-2}}(b_t + A_t \tilde{u}_t).} \tag{24}$$

Thus the implementation is weighted isotonic regression of $v_t^\alpha = b_t^\alpha + \sigma_t^\alpha \tilde{u}_t^\alpha$ with weights $w_t^\alpha = 1/(\sigma_t^\alpha)^2$.

The ordinary residual case has forecast-space metric I; the studentized case has metric A_t^{-2}. These weights follow from the chosen Euclidean geometry of the tracked score coordinates.

For $S_t^\alpha(y) = (y - b_t^\alpha)/\sigma_t$, we have $A_t = \sigma_t I$. Positive scalar multiplication preserves $\mathcal{K}$:

$$C_t^{\text{stu}} = \mathcal{K} - b_t/\sigma_t, \qquad q_t^{\text{stu}} = \Pi_{\mathcal{K}}(b_t + \sigma_t \tilde{u}_t). \tag{25}$$

All projection weights are equal and cancel. Equivalently, run standard MultiQT on base b_t/σ_t and outcome y_t/σ_t, then multiply the issued forecast by σ_t.

If $\sigma_t = \sigma$ is constant, standard MultiQT on the original inputs is also equivalent after converting the state and learning rate:

$$\tilde{\theta}_t = \sigma \tilde{u}_t, \qquad \eta_{\text{std}} = \sigma \eta_{\text{stu}}. \tag{26}$$

2 question and ideas

- What other scores would be reasonable to consider, i.e. what property of the base forecast do I need to recalibrate? Instead of asking for 1d PIT calibration, could I ask for 1d hdr-calibration? With studentized scores, or in general strictly increasing and invertible scores, the coverage indicator and calibration target do not change